SQL Command Types

Complete Beginner's Guide — DDL • DML • DQL • DCL • TCL

SQL statements are divided into **5 main categories**. Understanding these categories helps you know exactly what each command does and when to use it.

1. DDL — Data Definition Language

Purpose: Create or change the **structure** of the database (tables, columns, indexes, etc.). DDL changes the **design**, not the data itself.

Command	What it does	Example
CREATE	Creates a database or table	CREATE TABLE students (...);
ALTER	Changes table structure (add/modify column)	ALTER TABLE students ADD age INT;
DROP	Deletes a table or database permanently	DROP TABLE students;
TRUNCATE	Deletes all rows but keeps the table	TRUNCATE TABLE students;
RENAME	Renames a table	RENAME TABLE students TO learners;

Tip: DDL commands usually auto-commit (changes are permanent immediately).

2. DML — Data Manipulation Language

Purpose: Work with the **actual data** inside tables — insert new rows, update existing ones, or delete them. DML changes the **data**, not the structure.

Command	What it does	Example
INSERT	Adds new row(s) into a table	INSERT INTO students VALUES (1, 'Aisha');
UPDATE	Modifies existing data in a table	UPDATE students SET age = 21 WHERE id = 1;
DELETE	Removes row(s) from a table	DELETE FROM students WHERE id = 1;

Tip: Always use a WHERE clause with UPDATE and DELETE to avoid changing/deleting all rows!

3. DQL — Data Query Language

Purpose: Only used to **read / fetch** data from tables. It does not change anything. SELECT is the most frequently used SQL command.

Command	What it does	Example
SELECT	Retrieves data from one or more tables	SELECT * FROM students;
	Can filter, sort, and join data	SELECT name, age FROM students WHERE age > 18;

Note: Many people put SELECT under DML, but officially it belongs to DQL.

4. DCL — Data Control Language

Purpose: Control **who can access** the database and what they are allowed to do (security & permissions).

Command	What it does	Example
GRANT	Gives permission to a user	GRANT SELECT ON students TO rahul;
REVOKE	Removes permission from a user	REVOKE SELECT ON students FROM rahul;

5. TCL — Transaction Control Language

Purpose: Manage **transactions** — a group of SQL statements that must either all succeed or all fail together. Useful for keeping data consistent.

Command	What it does	Example
COMMIT	Saves all changes permanently	COMMIT;
ROLLBACK	Undoes changes made in the current transaction	ROLLBACK;
SAVEPOINT	Creates a temporary checkpoint inside a transaction	SAVEPOINT before_update;

Super Easy Memory Trick

Category	Full Form	Simple Meaning
DDL	Data Definition Language	Define / Design the tables
DML	Data Manipulation Language	Insert / Change the data
DQL	Data Query Language	Ask / Get the data
DCL	Data Control Language	Control access (who can do what)
TCL	Transaction Control Language	Control save / undo

Quick Classroom Quiz

1. Which category does **CREATE TABLE** belong to?
2. Which category does **UPDATE** belong to?
3. Which category does **SELECT** belong to?
4. Which category does **GRANT** belong to?
5. Which command would you use to undo a mistaken UPDATE?